Step 1: Probability Density and the Continuity Equation

The probability density of finding a quantum particle in a state described by the wave function $\Psi(\vec{r}, t)$ is: [MIT OpenCourseWare]

$$\rho = |\Psi|^2 = \Psi^* \Psi$$

In quantum mechanics, probability is conserved. This is expressed by the **continuity equation**:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$$

where $\vec{j}$ is the probability current density vector we want to find. [MIT OpenCourseWare]

Step 2: Time Derivative of Probability Density

Take the time derivative of $\rho = \Psi^* \Psi$:

$$\frac{\partial \rho}{\partial t} = \frac{\partial}{\partial t}(\Psi^* \Psi) = \Psi^* \frac{\partial \Psi}{\partial t} + \frac{\partial \Psi^*}{\partial t} \Psi$$

Now, use the **time-dependent Schrödinger equation** for a particle of mass m in a potential V:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi + V\Psi$$

Rearranging for $\frac{\partial \Psi}{\partial t}$:

$$\frac{\partial \Psi}{\partial t} = \frac{i\hbar}{2m} \nabla^2 \Psi - \frac{i}{\hbar} V\Psi$$

And take the complex conjugate for $\frac{\partial \Psi^*}{\partial t}$:

$$\frac{\partial \Psi^*}{\partial t} = -\frac{i\hbar}{2m} \nabla^2 \Psi^* + \frac{i}{\hbar} V\Psi^*$$

Step 3: Substitute into $\frac{\partial \rho}{\partial t}$

Substitute these expressions into the time derivative of ρ:

$$\frac{\partial \rho}{\partial t} = \Psi^* \left(\frac{i\hbar}{2m} \nabla^2 \Psi - \frac{i}{\hbar} V \Psi \right) + \Psi \left(-\frac{i\hbar}{2m} \nabla^2 \Psi^* + \frac{i}{\hbar} V \Psi^* \right)$$

Notice that the potential terms $V\Psi^*\Psi$ and $-V\Psi\Psi^*$ cancel out completely:

$$\frac{\partial \rho}{\partial t} = \frac{i\hbar}{2m} \left(\Psi^* \nabla^2 \Psi - \Psi \nabla^2 \Psi^* \right)$$

Step 4: Rewrite in Divergence Form

Notice that the term on the right can be written as the divergence of a vector:

$$\Psi^* \nabla^2 \Psi - \Psi \nabla^2 \Psi^* = \nabla \cdot (\Psi^* \nabla \Psi - \Psi \nabla \Psi^*)$$

Multiply by $\frac{i\hbar}{2m}$:

$$\frac{\partial \rho}{\partial t} = \nabla \cdot \left[\frac{i\hbar}{2m} (\Psi^* \nabla \Psi - \Psi \nabla \Psi^*) \right]$$

To match the continuity equation form $\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$, move the term to the left:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left[\frac{-i\hbar}{2m} (\Psi^* \nabla \Psi - \Psi \nabla \Psi^*) \right] = 0$$

Simplifying the factor of i (since $-i = \frac{1}{i}$ or distributing i properly):

$$\vec{j} = \frac{\hbar}{2mi} (\Psi^* \nabla \Psi - \Psi \nabla \Psi^*) = \frac{\hbar}{2mi} (\Psi^* \nabla \Psi - \Psi \nabla \Psi^*)$$

(Often written using the imaginary part):

$$\vec{j} = \frac{\hbar}{m} \operatorname{Im} (\Psi^* \nabla \Psi) = \frac{i\hbar}{2m} (\Psi \nabla \Psi^* - \Psi^* \nabla \Psi) \quad [0.6.1,\ 0.6.6]$$

Step 5: Transition to Electric Current Density

To get the electric current density ($\vec{J}_{\text{elec}}$), multiply the probability current density $\vec{j}$ by the electric charge q of the carrier (e.g., $q = -e$ for an electron): [Physics Stack Exchange]

$$\vec{J}_{\text{elec}} = q\vec{j} = -\frac{e\hbar}{2mi}\left(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*\right)$$

If a magnetic field is present (via minimal coupling, replacing $\nabla \to \nabla - \frac{iq}{\hbar}\vec{A}$), the electric current density becomes:

$$\vec{J}_{\text{elec}} = \frac{e\hbar}{2mi}\left(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*\right) - \frac{e^2}{m}\vec{A}|\Psi|^2$$

where $\vec{A}$ is the magnetic vector potential. [Kansas State University]